

Insulin Resistance and Systemic Inflammation Below the HbA1c Prediabetes Threshold: Age-Dependent Associations in U.S. Adults, NHANES 2015–March 2020

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Abstract

Background: Insulin resistance may be present despite glycemic measures remaining below conventional thresholds for prediabetes, but whether its relationship with systemic inflammation varies across adulthood is unclear. We examined the association between the Homeostatic Model Assessment of Insulin Resistance (HOMA-IR) and high-sensitivity C-reactive protein (hs-CRP) among adults with HbA1c < 5.7% and tested whether this association was modified by age.

Methods: We conducted a cross-sectional analysis of 3,079 adults aged ≥ 20 years without diagnosed diabetes and with HbA1c < 5.7% using harmonized pre-pandemic NHANES 2015–March 2020 data. Survey-weighted regression models estimated associations between HOMA-IR and log-transformed hs-CRP while adjusting for HbA1c, waist circumference, continuous age, sex, and non-HDL cholesterol. Effect modification was evaluated using a formal HOMA-IR-by-age-group interaction for ages 20–39, 40–54, and ≥ 55 years. Independent-cycle analyses, continuous-age modeling, and multiple sensitivity analyses evaluated reproducibility and robustness.

Results: The HOMA-IR–hs-CRP association differed across age groups ($F(2, 40) = 5.69$, $p = 0.00671$). A one-unit higher HOMA-IR was associated with approximately 2.7% higher hs-CRP among adults aged 20–39 years ($\beta = 0.0266$; 95% CI, 0.0038–0.0495) and 5.2% higher hs-CRP among those aged 40–54 years ($\beta = 0.0511$; 95% CI, 0.0132–0.0890). Among adults aged ≥ 55 years, the association was attenuated and not statistically distinguishable from zero ($\beta = -0.0189$; 95% CI, -0.0513 – 0.0135). Survey-weighted median HOMA-IR was similar across age groups, indicating that the interaction was not simply attributable to higher average HOMA-IR in midlife. The age-dependent pattern was broadly supported by independent-cycle analyses, continuous-age modeling, and sensitivity analyses, although the strength of formal interaction evidence varied across model specifications.

Conclusions: Among U.S. adults without diagnosed diabetes and with HbA1c below the conventional prediabetes threshold, the adjusted association between HOMA-IR and systemic

inflammation varied across age. These findings suggest that adults with HbA1c $< 5.7\%$ remain metabolically heterogeneous and that variation in insulin resistance may provide information about inflammatory physiology not captured by glycemic classification alone. Prospective studies are needed to determine whether measuring insulin resistance adds clinically useful information to conventional glycemic assessment and whether such utility varies with age.

1 Introduction

Glycohemoglobin (HbA1c) plays a central role in the identification of dysglycemia and diabetes risk. Because it reflects cumulative glycemic exposure over preceding weeks to months [Nathan et al., 2007], HbA1c provides clinically useful information about glucose regulation and is incorporated into widely used diagnostic thresholds for prediabetes and diabetes [American Diabetes Association Professional Practice Committee, 2026]. Glycemic status, however, is not synonymous with insulin sensitivity. During the development of insulin resistance, compensatory increases in insulin secretion may preserve circulating glucose concentrations despite diminished tissue responsiveness to insulin [Rodzeń et al., 2026]. Individuals who remain below conventional HbA1c thresholds may therefore occupy different metabolic states despite sharing the same glycemic classification.

The homeostatic model assessment of insulin resistance (HOMA-IR), derived from fasting glucose and fasting insulin concentrations, provides a surrogate estimate of insulin resistance and captures information not contained in HbA1c alone [Matthews et al., 1985, Wallace et al., 2004]. Although HOMA-IR is not a direct physiological measurement of insulin sensitivity, it is well suited to large population studies in which more intensive methods such as the hyperinsulinemic-euglycemic clamp are not feasible [DeFronzo et al., 1979, Wallace et al., 2004]. The coexistence of relatively normal glycemia with higher fasting insulin and HOMA-IR therefore provides one plausible form of metabolic heterogeneity beneath conventional glycemic thresholds.

Insulin resistance is also embedded within a broader network of metabolic and inflammatory processes. Adipose tissue dysfunction, altered lipid handling, hepatic metabolism, and immune signaling are closely interconnected with insulin action, while inflammatory pathways can themselves influence insulin signaling [Park et al., 2014, Chen et al., 2004, Park et al., 2009, Yan et al., 2019]. High-sensitivity C-reactive protein (hs-CRP), an acute-phase protein produced primarily by the liver in response to inflammatory signaling, provides a nonspecific marker of systemic inflammatory activity [Pepys and Hirschfield, 2003]. An association between HOMA-IR and hs-CRP would not establish a causal relationship between insulin resistance and inflammation, but could indicate that variation in insulin resistance is associated with physiologically relevant inflammatory heterogeneity.

Previous population-based studies have established associations between insulin resistance and systemic inflammatory markers in adults without diabetes. Analyses of U.S. NHANES data and other population cohorts have reported positive associations between HOMA-IR or fasting insulin

and C-reactive protein after adjustment for demographic, anthropometric, and metabolic covariates [Chen et al., 2004, Meng et al., 2007, Park et al., 2009, Missel et al., 2021]. Related work has demonstrated metabolic and inflammatory heterogeneity even within populations without overt diabetes or conventional hyperglycemia [Caporaso et al., 2020].

More recently, Shahid et al. [2023] examined HOMA-IR and C-reactive protein in a nationally representative Canadian population without diabetes and considered potential heterogeneity by sex and age. However, that study included participants with HbA1c values extending through the conventional prediabetes range ($< 6.5\%$), whereas the present study restricts the primary population to HbA1c $< 5.7\%$. Its principal findings also centered on sex rather than the age-dependent association examined here. Similarly, Caporaso et al. [2020] examined HOMA-IR across demographic and metabolic strata in a broader NHANES population but did not focus on formal age modification of the HOMA-IR–CRP relationship among adults restricted to HbA1c below 5.7%.

Collectively, these studies establish that the relationship between insulin resistance and inflammation is not itself a new observation. However, comparatively little population-based evidence has addressed whether age modifies the HOMA-IR–hs-CRP association specifically among adults who remain below the conventional HbA1c prediabetes threshold of 5.7%.

This question is relevant to recent calls for improved recognition and characterization of insulin resistance before the development of overt metabolic disease. Rodzeń et al. [2026], in a recent narrative review, highlighted the absence of standardized clinical criteria for insulin resistance and called for improved diagnostic frameworks and further research to support earlier identification. Importantly, they also emphasized the heterogeneity of insulin resistance and the need to interpret surrogate measures within appropriate clinical and physiological contexts. Their framework raises an empirical question relevant to glycemic classification: among individuals who remain below conventional glycemic thresholds, does a measure of insulin resistance provide biologically relevant information not captured by glycemic classification alone?

In this context, we use the term metabolic screening “blind spot” in a limited sense. The term does not imply that HbA1c is an inadequate measure of glycemia, that individuals with HbA1c below 5.7% are necessarily metabolically unhealthy, or that HOMA-IR should replace established screening measures. Rather, it refers to the possibility that classification based on glycemia alone may leave meaningful variation in insulin resistance unresolved among individuals assigned to the same glycemic category. If that variation is systematically associated with inflammatory status, then glycemic classification and insulin-resistance measures may provide complementary information about metabolic phenotype.

The National Health and Nutrition Examination Survey (NHANES) provides an opportunity to examine this question in a nationally representative U.S. population with simultaneous measurements of HbA1c, fasting glucose, fasting insulin, anthropometry, lipids, and hs-CRP [Akinbami et al., 2022]. In the present study, we used harmonized pre-pandemic NHANES data from 2015 through

March 2020 to examine adults aged 20 years or older without diagnosed diabetes and with HbA1c below 5.7%. The original study hypothesis was that HOMA-IR would be associated with systemic inflammatory status within this population, consistent with metabolic heterogeneity not captured by glycemic classification alone.

During methodological evaluation of the initially observed age-stratified associations, age emerged as a potential effect modifier. The revised analysis therefore formally evaluated the HOMA-IR-by-age interaction using a pooled survey-weighted model rather than inferring effect modification from differences in statistical significance across separately fitted age strata. The age-dependent pattern was subsequently evaluated using continuous nonlinear modeling of age, independent-cycle replication, alternative outcome and exposure transformations, anthropometric specifications, inflammatory restrictions, and additional sensitivity analyses. Accordingly, the present study addresses both the original question of whether HOMA-IR is associated with inflammatory heterogeneity below the conventional HbA1c prediabetes threshold and the subsequently identified question of whether that association varies across adulthood.

2 Methods

2.1 Study Design and Data Source

We conducted a cross-sectional analysis of publicly available data from the National Health and Nutrition Examination Survey (NHANES), using the 2015–2016 cycle and the 2017–March 2020 pre-pandemic data release. NHANES is a nationally representative survey of the noninstitutionalized U.S. population that combines interviews, physical examinations, and laboratory measurements using a complex, multistage probability sampling design [[National Center for Health Statistics, 2021a](#)].

The 2015–2016 and 2017–March 2020 datasets were harmonized at the participant level using the NHANES participant identifier (SEQN). Variables required for the present analysis were obtained from the demographic, examination, diabetes questionnaire, glycohemoglobin, fasting glucose, fasting insulin, high-sensitivity C-reactive protein, and lipid files. Because HOMA-IR requires both fasting glucose and fasting insulin, analyses were conducted within the NHANES fasting laboratory subsample and used the corresponding fasting-subsample survey weights.

The primary analysis combined the available pre-pandemic observations from 2015 through March 2020. The two conventional 2015–2016 and 2017–2018 NHANES cycles were additionally analyzed separately as an independent-cycle replication analysis [[National Center for Health Statistics, 2021a](#)].

2.2 Analytic Population

The analytic population was restricted to adults aged 20 years or older who had no self-reported diagnosis of diabetes and had $\text{HbA1c} < 5.7\%$, the lower boundary of the conventional HbA1c-defined prediabetes range [American Diabetes Association Professional Practice Committee, 2026]. Participants were additionally required to belong to the qualifying NHANES fasting laboratory subsample, as indicated by a nonzero fasting-subsample survey weight, and to have valid fasting glucose and fasting insulin measurements, positive HOMA-IR and hs-CRP values, and complete data for variables included in the primary harmonized model. No imputation of missing data was performed.

Fasting-subsample eligibility was independently verified using participant-reported fasting duration from the NHANES fasting questionnaire, calculated as $\text{PHAFSTHR} + \text{PHAFSTMN}/60$ [National Center for Health Statistics, 2017, 2021b]. All participants with a qualifying fasting-subsample weight in the final analytic population reported fasting durations within the NHANES qualifying range of 8 to < 24 hours, confirming concordance between fasting-subsample eligibility and reported fasting duration.

Of 3,182 participants meeting the remaining eligibility and measurement requirements, 103 (3.2%) were excluded because of incomplete primary-model covariate data, yielding a final primary analytic sample of 3,079 participants (Supplementary Figure S1). The final sample represented approximately 144.3 million U.S. adults after application of the combined survey weights.

No a priori sample-size calculation was performed; the analytic sample comprised all participants from the specified NHANES survey periods who met the study eligibility criteria and had the data required for the primary analysis.

The HbA1c restriction was used to examine metabolic heterogeneity specifically among individuals remaining below the conventional laboratory threshold for prediabetes. This restriction defines the population under study and should not be interpreted as establishing that all included participants were otherwise metabolically healthy.

2.3 Exposure: HOMA-IR

The primary exposure was the Homeostatic Model Assessment of Insulin Resistance (HOMA-IR), a surrogate measure of insulin resistance derived from fasting insulin and fasting glucose concentrations [Matthews et al., 1985, Wallace et al., 2004]. HOMA-IR was calculated as

$$\text{HOMA-IR} = \frac{\text{fasting insulin } (\mu\text{U/mL}) \times \text{fasting glucose (mg/dL)}}{405}.$$

HOMA-IR was modeled continuously and retained on its original scale in the primary analysis.

Although HOMA-IR was right-skewed, log transformation of the exposure did not materially improve residual behavior when hs-CRP was modeled on the log scale. Retaining HOMA-IR on its original scale also permits direct interpretation of model coefficients per one-unit difference in HOMA-IR. Log-transformed HOMA-IR was evaluated in sensitivity analyses.

2.4 Outcome: hs-CRP

The primary outcome was high-sensitivity C-reactive protein (hs-CRP), modeled as its natural logarithm,

$$Y = \ln(\text{hsCRP}).$$

Raw hs-CRP exhibited substantial right skew and produced strongly skewed and heavy-tailed regression residuals. Log transformation markedly improved residual behavior and was therefore selected for the primary model on statistical-distributional grounds rather than on the basis of statistical significance.

For models using $\ln(\text{hsCRP})$, coefficients were converted when useful to proportional differences on the original hs-CRP scale according to

$$100 \left(e^{\beta} - 1 \right) \%.$$

Models using untransformed hs-CRP were retained as sensitivity analyses.

2.5 Age and Covariates

Following the methodological evaluation of the initial age-stratified findings, age was designated as the primary effect modifier in the revised analysis. For the principal analysis and presentation, participants were classified into three age groups:

$$20\text{--}39, \quad 40\text{--}54, \quad \geq 55 \text{ years}.$$

Age was also retained as a continuous covariate within the adjusted model to account for within-category age variation. The categorical age groups were selected to provide interpretable estimates of effect modification; sensitivity analyses treated age continuously and allowed the HOMA-IR association to vary nonlinearly across adulthood.

The primary harmonized adjustment set consisted of HbA1c, waist circumference, continuous age,

sex, and non-HDL cholesterol. Non-HDL cholesterol was calculated as

$$\text{non-HDL} = \text{total cholesterol} - \text{HDL cholesterol}.$$

Waist circumference was selected as the primary anthropometric covariate because it was consistently available across the survey periods included in the combined analysis and provides a measure of central adiposity. This choice was also consistent with NHLBI guidance identifying waist circumference as a practical anthropometric measure of abdominal adiposity and a better marker of abdominal fat content than waist-to-hip ratio [NHLBI Obesity Education Initiative Expert Panel on the Identification, Evaluation, and Treatment of Obesity in Adults, 1998]. Waist-to-hip ratio was not used in the primary combined-period model because hip circumference was not available consistently across all included survey periods.

2.6 NHANES Survey Design and Weighting

All population estimates and inferential analyses incorporated the NHANES complex sampling design [National Center for Health Statistics, 2021a]. Because calculation of HOMA-IR required both fasting glucose and fasting insulin, analyses were conducted within the qualifying NHANES fasting laboratory subsample and used the corresponding fasting-subsample survey weights rather than general examination weights.

Fasting-subsample eligibility was independently verified using participant-reported fasting duration from the NHANES fasting questionnaire, calculated as $\text{PHAFSTHR} + \text{PHAFSTMN}/60$. All participants with a qualifying fasting-subsample weight in the final analytic population reported fasting durations within the NHANES qualifying range of 8 to < 24 hours, confirming concordance between fasting-subsample eligibility and reported fasting duration.

Survey variance estimation additionally incorporated the NHANES masked variance stratum (SDMVSTRA) and masked variance unit (SDMVPSU) variables [National Center for Health Statistics, 2021a]. These variables account for the stratified and clustered public-use survey design and were used for variance estimation rather than as biological covariates.

For the combined 2015–March 2020 analysis, fasting weights were adjusted for the unequal durations represented by the component datasets according to the NHANES guidance [National Center for Health Statistics, 2021a]. The 2015–2016 fasting weight was rescaled as

$$W_{15/16}^* = \text{WTSF2YR} \left(\frac{2}{5.2} \right),$$

and the 2017–March 2020 pre-pandemic fasting weight as

$$W_{17/20}^* = \text{WTSAFP RP} \left(\frac{3.2}{5.2} \right).$$

Masked variance strata and variance units were uniquely identified across periods before the datasets were combined. The resulting combined design provided 40 survey design degrees of freedom.

Survey-design-based variance estimates were used for standard errors, confidence intervals, t -tests, and F -tests. Participants were therefore not treated as independent simple-random-sample observations.

2.7 Primary Statistical Analysis

The primary analysis estimated the adjusted association between HOMA-IR and log-transformed hs-CRP and formally tested whether that association differed across age groups. The model can be represented as

$$\ln(\text{hsCRP}_i) = \beta_0 + \beta_1 H_i + \beta_2 A_i + \beta_3 (H_i \times A_i) + \gamma^T Z_i + \epsilon_i,$$

where H_i denotes HOMA-IR, A_i denotes age group, and Z_i contains HbA1c, waist circumference, continuous age, sex, and non-HDL cholesterol.

The primary inferential test was the joint survey-adjusted test of

$$H_0 : \beta_{\text{HOMA},20-39} = \beta_{\text{HOMA},40-54} = \beta_{\text{HOMA},55+}.$$

Thus, evidence for effect modification was determined from the HOMA-IR-by-age interaction itself rather than by comparing whether independently estimated age-specific coefficients crossed a conventional significance threshold.

Age-specific HOMA-IR slopes and their 95% confidence intervals were subsequently estimated from the fitted interaction model to characterize the direction and magnitude of the association within each age group.

For comparison with the primary adjusted estimates, unadjusted age-specific HOMA-IR associations were also estimated in the same complete-case analytic population using the survey-weighted HOMA-IR-by-age-group model without the primary adjustment covariates; survey period was retained as a nuisance term in the combined-period analysis.

For the combined-period analysis, a survey-period indicator was included, and the adjustment covariates—HbA1c, waist circumference, continuous age, sex, and non-HDL cholesterol—were permitted to have period-specific coefficients through interactions with survey period. The HOMA-

IR-by-age-group association remained the common effect-modification structure of primary interest. This specification avoided unnecessarily assuming that nuisance covariate relationships were identical in the 2015–2016 and 2017–March 2020 components.

Several design and analytic features were used to reduce potential bias. NHANES fasting-subsample survey weights were applied to account for the complex sampling design and differential probability of selection into the fasting subsample. Eligibility and exposure definitions were applied consistently across survey periods, and the primary model used harmonized covariates available across the full study period. Sensitivity and replication analyses were used to evaluate the robustness of the observed age-dependent association to alternative model specifications, outcome and exposure scales, survey periods, and additional potential confounders. As an independent cross-software validation, the primary analysis and selected sensitivity analyses were reimplemented from the raw NHANES files in R using the survey package. Analytic sample construction, survey design degrees of freedom, model estimates, age-specific contrasts, and joint Wald statistics reproduced the primary Python implementation to numerical precision.

2.8 Independent-Cycle Replication

To assess reproducibility independently of the expanded combined analysis, harmonized models were fitted separately to the conventional 2015–2016 and 2017–2018 NHANES cycles. Within each cycle, the HOMA-IR-by-age interaction and corresponding age-specific slopes were estimated using the appropriate fasting-subsample weights and survey-design variables.

Replication was evaluated from the consistency of estimated age-dependent patterns and by a formal HOMA-IR-by-age-group-by-cycle interaction testing whether age-specific HOMA-IR associations differed between cycles. Statistical significance of an age-specific coefficient in one cycle and nonsignificance in another was not, by itself, considered evidence of between-cycle heterogeneity.

2.9 Continuous-Age Analysis

To evaluate whether the categorical age findings depended on the selected age boundaries, age was additionally modeled continuously. We examined linear and quadratic HOMA-IR-by-age interactions and a natural cubic spline representation of age using four degrees of freedom.

The spline model estimated the adjusted HOMA-IR slope as a continuous function of age,

$$\beta_{\text{HOMA}}(a) = \frac{\partial E[\ln(\text{hsCRP}) \mid X]}{\partial \text{HOMA-IR}},$$

where a denotes age.

This analysis was used to assess the shape of age modification rather than to identify a discrete biological age threshold. In particular, crossing of the estimated slope or its confidence interval through zero was not interpreted as defining a specific age at which the HOMA-IR–hs-CRP association begins or ends.

2.10 Sensitivity Analyses

Sensitivity analyses were organized according to the potential source of inferential vulnerability they were intended to address rather than treated as additional primary analyses.

Because HbA1c was used both to define the analytic population ($\text{HbA1c} < 5.7\%$) and as a continuous adjustment covariate in the primary model, we additionally repeated the primary analysis without continuous HbA1c adjustment while retaining the HbA1c eligibility criterion. This analysis assessed whether conditioning on residual within-range variation in HbA1c materially influenced the estimated HOMA-IR–hs-CRP association or its modification by age.

Outcome distribution and inflammatory extremes. We repeated analyses using untransformed hs-CRP and examined models restricting hs-CRP to ≤ 10 mg/L to assess sensitivity to observations potentially reflecting acute inflammatory states. The inflammatory-restriction analysis was evaluated using both log-transformed and untransformed hs-CRP outcomes.

Exposure scale. We evaluated log-transformed HOMA-IR in combination with both raw and log-transformed hs-CRP. Together with the primary specification, these analyses formed a four-model transformation grid evaluating raw and log scales for both exposure and outcome.

Age specification. In addition to the primary categorical interaction, HOMA-IR effect modification was evaluated using continuous linear age, quadratic age, and a four-degree-of-freedom natural cubic spline representation.

Adiposity specification. Because adiposity is closely related to both insulin resistance and systemic inflammation, the dependence of the primary findings on the anthropometric specification was examined explicitly. Waist circumference remained the harmonized primary adiposity measure. The primary model was refitted using BMI and waist-to-height ratio in place of waist circumference. These comparisons were additionally performed in a common analytic sample with complete data for all three measures to distinguish differences due to anthropometric specification from differences in sample composition. Waist-to-hip ratio was evaluated separately during the survey period in which hip circumference was available, with comparisons among waist circumference, BMI, waist-to-height ratio, and waist-to-hip ratio restricted to the same participants.

HbA1c eligibility. Additional sensitivity analyses progressively relaxed the HbA1c eligibility criterion from $< 5.7\%$ to $< 6.0\%$, $< 6.5\%$, and no HbA1c restriction while continuing to exclude participants with diagnosed diabetes. These analyses were used to assess whether the observed age-dependent pattern, particularly attenuation among older adults, could be explained primarily by the original HbA1c selection threshold.

Survey-cluster and influence analyses. Survey-cluster influence was evaluated using a leave-one-cluster-out procedure in which each unique masked stratum–PSU combination was removed in turn and the primary model refitted. Additional influence analyses examined the sensitivity of results to influential HOMA-IR and hs-CRP observations and to restrictions on extreme values.

Additional supportive analyses. Where the required variables were available and scientifically appropriate, additional analyses evaluated smoking, medication burden, statin use, renal and liver/metabolic biochemical variables, expanded covariate sets, and sex heterogeneity. These analyses were considered supportive or exploratory rather than additional primary hypothesis tests and were not uniformly available across the full harmonized study period.

2.11 Descriptive Statistics and Statistical Inference

Characteristics of the primary analytic population were summarized overall and by age group. Approximately symmetric continuous variables were reported as survey-weighted means with standard errors. Because fasting insulin, HOMA-IR, and hs-CRP were strongly right-skewed, these variables were summarized using survey-weighted medians and interquartile ranges.

Regression coefficients are reported with survey-adjusted standard errors, 95% confidence intervals, and two-sided p -values where appropriate. Joint interaction hypotheses were evaluated using survey-adjusted F -tests. A conventional two-sided $\alpha = 0.05$ threshold was used as a reference for statistical inference; however, interpretation considered effect magnitude, confidence intervals, consistency across analyses, and the distinction between primary and sensitivity analyses rather than treating $p = .05$ as a strict boundary between the presence and absence of an association.

2.12 Statistical Software and Computational Implementation

Data management and statistical analyses were performed in Python (version 3.13.5). Data manipulation was conducted using pandas (version 2.2.3), numerical matrix operations using NumPy (version 2.3.5), regression design matrices using patsy (version 1.0.2), and statistical distribution functions using SciPy (version 1.17.0).

Survey-weighted regression coefficients were estimated directly using weighted least squares. For

a design matrix $\mathbf{X}$, outcome vector $\mathbf{y}$, and diagonal matrix of survey weights $\mathbf{W}$, the coefficient estimator was

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}.$$

The survey weights determined the contribution of each observation to the coefficient estimates, whereas uncertainty estimation additionally incorporated the NHANES masked variance strata and primary sampling units. Weighted regression score contributions were aggregated within each masked primary sampling unit and centered within the corresponding masked variance stratum. The resulting stratum-level score variation was used to construct a design-based sandwich covariance estimator,

$$\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) = \mathbf{B}^{-1} \mathbf{M} \mathbf{B}^{-1},$$

where

$$\mathbf{B} = \mathbf{X}^T \mathbf{W} \mathbf{X}$$

is the weighted regression information matrix and $\mathbf{M}$ represents the covariance of PSU-level score contributions accumulated across survey strata. Within each stratum containing m_h sampled PSUs, the contribution to $\mathbf{M}$ incorporated the finite-PSU correction factor $m_h/(m_h - 1)$.

Survey design degrees of freedom were calculated as the number of contributing masked primary sampling units minus the number of contributing masked variance strata. Wald statistics based on the resulting design-based covariance matrix were used for coefficient tests and joint tests of interaction parameters. Joint hypotheses involving multiple interaction coefficients were evaluated using survey-adjusted F -statistics with numerator degrees of freedom equal to the number of independent restrictions and denominator degrees of freedom determined by the survey design.

Model formulas and interaction terms were represented explicitly in the regression design matrix. Thus, the HOMA-IR-by-age-group analysis was fit as a single pooled survey-weighted regression containing the appropriate main-effect and product-term columns, rather than by comparing statistical significance across separately fitted subgroup regressions.

All primary, replication, and sensitivity analyses were implemented using the same computational framework, with analysis-specific changes limited to the population definition, exposure or outcome transformation, covariate specification, or interaction structure described above.

3 Results

3.1 Study Population

The final 2015–March 2020 analytic sample included 3,079 adults meeting the prespecified eligibility criteria and represented approximately 144.3 million U.S. adults after application of the combined fasting-subsample survey weights. The analytic sample included 1,436 participants aged 20–39 years, 775 aged 40–54 years, and 868 aged 55 years or older.

Survey-weighted characteristics of the analytic population are shown in Table 1. Overall, the weighted median HOMA-IR was 1.95 (interquartile range [IQR], 1.23–3.20), and the weighted median hs-CRP was 1.47 mg/L (IQR, 0.64–3.23).

Table 1: Survey-weighted characteristics of the NHANES 2015–March 2020 analytic population

Characteristic	Overall	20–39	40–54	55+
Unweighted n	3,079	1,436	775	868
Weighted population, millions	144.3	70.5	36.2	37.7
Age, years	42.8 (0.5)	29.0 (0.2)	46.7 (0.2)	65.1 (0.3)
Male, %	48.6 (1.2)	49.6 (1.6)	50.9 (2.4)	44.6 (2.0)
HbA1c, %	5.25 (0.01)	5.17 (0.01)	5.29 (0.01)	5.37 (0.01)
Fasting glucose, mg/dL	99.4 (0.3)	97.1 (0.4)	100.4 (0.6)	102.6 (0.5)
Waist circumference, cm	96.8 (0.5)	95.1 (0.8)	97.4 (0.9)	99.3 (0.7)
BMI, kg/m ²	28.3 (0.2)	28.3 (0.3)	28.4 (0.3)	27.9 (0.3)
Total cholesterol, mg/dL	187.9 (1.1)	176.3 (1.5)	199.3 (2.3)	198.5 (2.3)
HDL cholesterol, mg/dL	57.2 (0.5)	54.3 (0.6)	56.4 (0.9)	63.3 (1.1)
Non-HDL cholesterol, mg/dL	130.7 (1.1)	122.0 (1.4)	142.9 (2.1)	135.2 (2.3)
Fasting insulin, uU/mL	7.83 [5.12, 12.54]	8.16 [5.06, 13.41]	7.62 [4.96, 11.86]	7.59 [5.31, 11.74]
HOMA-IR	1.95 [1.23, 3.20]	1.99 [1.20, 3.24]	1.94 [1.19, 3.10]	1.89 [1.32, 3.15]
hs-CRP, mg/L	1.47 [0.64, 3.23]	1.39 [0.57, 3.30]	1.44 [0.70, 3.15]	1.59 [0.80, 3.37]

Notes: Values are survey-weighted mean (standard error) unless otherwise indicated. Fasting insulin, HOMA-IR, and hs-CRP are reported as survey-weighted median [interquartile range] because of substantial right skew. Male is reported as weighted percentage (standard error). The analytic population includes adults aged ≥ 20 years with no diagnosed diabetes, HbA1c $< 5.7\%$, a qualifying fasting-subsample weight, and complete variables required for the primary harmonized model. Combined weights use the 2015–2016 fasting weight multiplied by 2/5.2 and the 2017–March 2020 pre-pandemic fasting weight multiplied by 3.2/5.2.

Survey-weighted median HOMA-IR was similar across age groups: 1.99 (IQR, 1.20–3.24) among adults aged 20–39 years, 1.94 (1.19–3.10) among those aged 40–54 years, and 1.89 (1.32–3.15) among those aged 55 years or older. Median hs-CRP was modestly higher among older adults. These descriptive comparisons were not used to infer age modification of the adjusted HOMA-IR–hs-CRP association.

3.2 Primary HOMA-IR–hs-CRP Association and Modification by Age

In the primary survey-adjusted model with log-transformed hs-CRP as the outcome, there was evidence that the adjusted association between HOMA-IR and hs-CRP differed across age groups. The joint test of the HOMA-IR-by-age-group interaction yielded

$$F(2, 40) = 5.69, \quad p = 0.00671.$$

Age-specific estimates from the interaction model are shown in Figure 1. Among adults aged 20–39 years, the HOMA-IR coefficient was

$$\beta = 0.0266 \quad (95\% \text{ CI, } 0.0038 \text{ to } 0.0495; p = 0.023).$$

Among adults aged 40–54 years, the corresponding coefficient was

$$\beta = 0.0511 \quad (95\% \text{ CI, } 0.0132 \text{ to } 0.0890; p = 0.0094).$$

Among adults aged 55 years or older, the coefficient was

$$\beta = -0.0189 \quad (95\% \text{ CI, } -0.0513 \text{ to } 0.0135; p = 0.246).$$

In unadjusted analyses, HOMA-IR was positively associated with hs-CRP across all three age groups. The age-specific coefficients were 0.1308 (95% CI, 0.0919–0.1697) among adults aged 20–39 years, 0.1553 (95% CI, 0.0982–0.2125) among adults aged 40–54 years, and 0.1017 (95% CI, 0.0512–0.1523) among adults aged ≥ 55 years. Evidence for age modification was not statistically significant in the unadjusted model ($F(2, 40) = 1.71$, $p = 0.194$). After adjustment for HbA1c, waist circumference, continuous age, sex, and non-HDL cholesterol, the association was substantially attenuated, particularly among older adults, producing the age-dependent pattern observed in the primary model.

Because the outcome was modeled on the natural-log scale, the coefficients can also be expressed as proportional differences in hs-CRP. A one-unit higher HOMA-IR was associated with approximately

$$100(e^{0.0266} - 1)\% = 2.7\%$$

higher hs-CRP among adults aged 20–39 years and approximately

$$100(e^{0.0511} - 1)\% = 5.2\%$$

higher hs-CRP among adults aged 40–54 years, conditional on the covariates included in the model. The estimate among adults aged 55 years or older was not statistically distinguishable from zero.

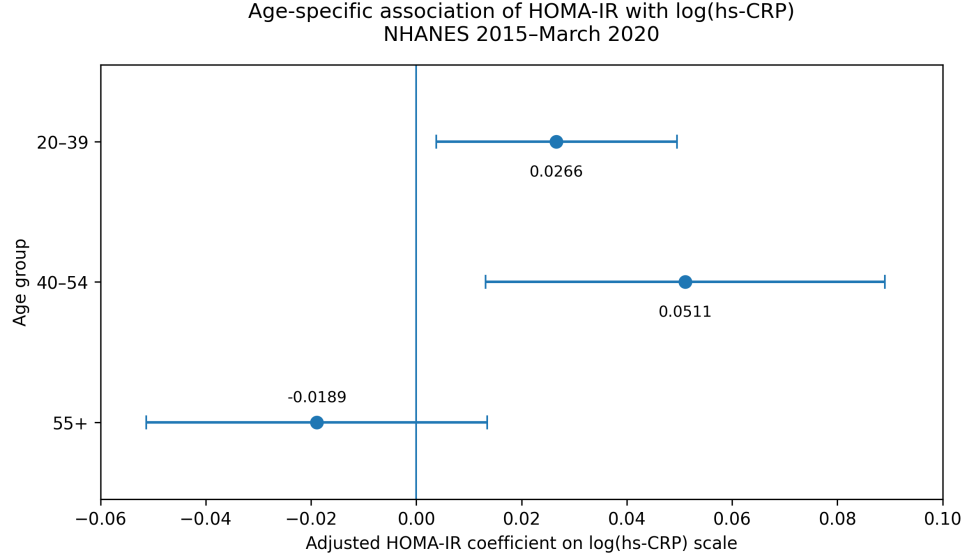


Figure 1: Adjusted age-specific associations between HOMA-IR and log-transformed hs-CRP in U.S. adults with HbA1c < 5.7%, NHANES 2015–March 2020. Points represent estimated HOMA-IR coefficients and horizontal bars represent 95% confidence intervals. Estimates were obtained from the primary survey-adjusted interaction model controlling for HbA1c, waist circumference, continuous age, sex, and non-HDL cholesterol. The vertical reference line denotes a coefficient of zero.

3.3 Continuous-Age Analysis

Continuous-age analyses were performed to determine whether the observed effect modification depended on the prespecified age-group boundaries. A model imposing a linear HOMA-IR-by-age interaction provided borderline evidence of age modification,

$$F(1, 40) = 3.93, \quad p = 0.054.$$

Allowing the relationship to vary quadratically strengthened the evidence for age modification,

$$F(2, 40) = 3.43, \quad p = 0.042,$$

although the additional quadratic interaction component considered separately did not provide strong evidence for a specifically quadratic functional form ($p = 0.086$).

A four-degree-of-freedom natural cubic spline model also supported age-dependent variation in the HOMA-IR association,

$$F(4, 40) = 3.09, \quad p = 0.026.$$

As shown in Figure 2, the estimated HOMA-IR coefficient remained positive through early and middle adulthood and subsequently attenuated with increasing age. The spline was used to characterize the continuous shape of the interaction rather than to identify a specific age threshold.

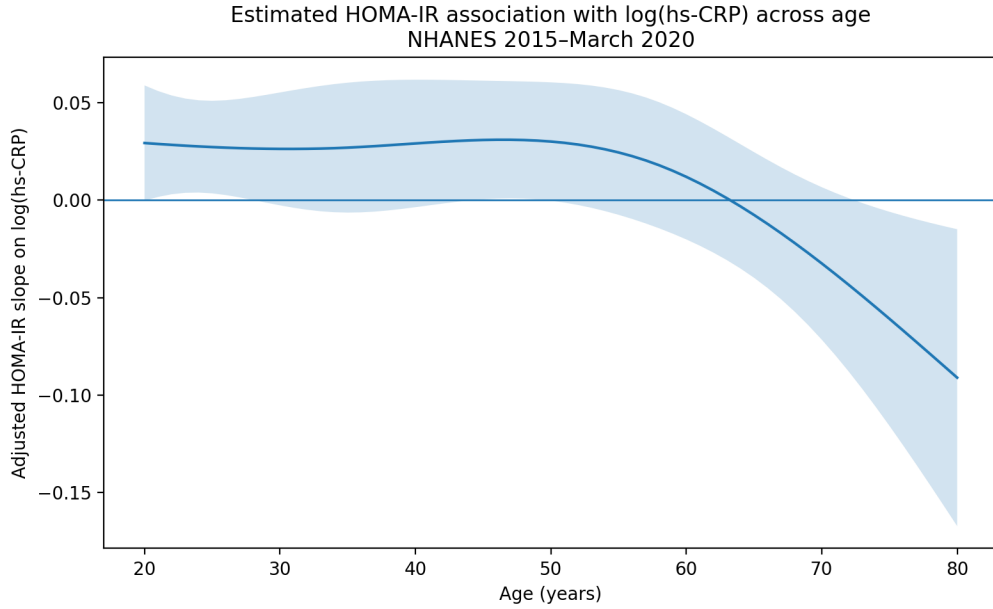


Figure 2: Estimated age-varying association between HOMA-IR and log-transformed hs-CRP in U.S. adults with $\text{HbA1c} < 5.7\%$, NHANES 2015–March 2020. The solid curve represents the estimated adjusted HOMA-IR coefficient as a continuous function of age, and the shaded region represents the 95% confidence interval. The horizontal reference line denotes no association. The curve represents variation in the HOMA-IR–hs-CRP association with age and does not represent the age trajectory of hs-CRP itself.

3.4 Independent-Cycle Replication

The age-dependent association was additionally evaluated in the independent 2015–2016 and 2017–2018 NHANES cycles using harmonized waist-adjusted models. On the raw hs-CRP scale, the formal HOMA-IR-by-age interaction was significant in both 2015–2016,

$$F(2, 15) = 4.48, \quad p = 0.0298,$$

and 2017–2018,

$$F(2, 15) = 7.26, \quad p = 0.00625.$$

The raw-scale HOMA-IR coefficients for the 20–39, 40–54, and 55+ year age groups were, respectively, 0.114, 0.329, and -0.028 in 2015–2016 and 0.298, 0.229, and -0.341 in 2017–2018.

A formal HOMA-IR-by-age-group-by-cycle interaction did not provide strong evidence that the age-dependent pattern differed between the two cycles,

$$F(2, 30) = 2.22, \quad p = 0.126.$$

The replication analysis was also repeated using log-transformed hs-CRP. The age-specific log-scale HOMA-IR coefficients for 2015–2016 and 2017–2018, respectively, were 0.025 and 0.036 among adults aged 20–39 years, 0.074 and 0.057 among adults aged 40–54 years, and -0.030 and -0.063 among adults aged 55 years or older.

On the log scale, the formal cross-cycle heterogeneity test yielded

$$F(2, 30) = 0.586, \quad p = 0.563.$$

The corresponding direct cross-cycle comparisons gave $p = 0.741$, $p = 0.659$, and $p = 0.373$ for the young, midlife, and older age groups, respectively. Thus, the data did not provide evidence that the age-specific HOMA-IR associations differed between the two independent survey cycles.

3.5 Sensitivity and Robustness Analyses

The primary finding was evaluated across alternative outcome and exposure scales, restrictions on inflammatory extremes, alternative adiposity adjustment, alternative HbA1c eligibility criteria, continuous-age specifications, and survey-cluster influence. Detailed results are presented in Supplementary Table S1.

When hs-CRP was analyzed on its original scale in the combined 2015–March 2020 population, evidence of age modification remained,

$$F(2, 40) = 3.96, \quad p = 0.0269.$$

The corresponding raw-scale HOMA-IR coefficients were 0.127 (95% CI, 0.018 to 0.235) among adults aged 20–39 years, 0.160 (95% CI, -0.056 to 0.377) among adults aged 40–54 years, and

−0.109 (95% CI, −0.282 to 0.064) among adults aged 55 years or older.

Evidence for interaction was less pronounced when HOMA-IR was log-transformed. With log-transformed hs-CRP as the outcome, the HOMA-IR-by-age interaction yielded

$$F(2, 40) = 1.86, \quad p = 0.169,$$

and when log-transformed HOMA-IR was combined with untransformed hs-CRP, the interaction yielded

$$F(2, 40) = 1.43, \quad p = 0.251.$$

Thus, the qualitative age-specific pattern remained recognizable across exposure and outcome transformations, but the strength of formal interaction evidence varied across scale specifications.

Restriction of hs-CRP to values ≤ 10 mg/L produced a similar distinction between outcome scales. With untransformed hs-CRP, evidence of age modification remained,

$$F(2, 40) = 4.03, \quad p = 0.0255,$$

with age-specific HOMA-IR coefficients of 0.0824, 0.0905, and −0.0304 among adults aged 20–39, 40–54, and 55 years or older, respectively. With log-transformed hs-CRP under the same restriction, the corresponding interaction was attenuated,

$$F(2, 40) = 2.58, \quad p = 0.0883.$$

Alternative adiposity specifications. The age-dependent HOMA-IR–hs-CRP association was not specific to the use of waist circumference as the primary adiposity covariate. In a common analytic sample with complete waist circumference, BMI, and waist-to-height ratio data ($n = 3,072$), the HOMA-IR-by-age interaction remained significant when the primary model was adjusted for waist circumference ($p = 0.0066$), BMI ($p = 0.0124$), or waist-to-height ratio ($p = 0.0023$). Across all three specifications, the adjusted HOMA-IR association remained positive among adults aged 20–39 and 40–54 years and was substantially attenuated among adults aged ≥ 55 years.

Waist-to-hip ratio could be evaluated only during the survey period in which hip circumference was available. Within a common sample from that period ($n = 1,899$), formal evidence for age modification was weaker across adiposity specifications, including waist circumference. Adjustment for waist-to-hip ratio produced less attenuation of the HOMA-IR association among adults aged ≥ 55 years than did adjustment for waist circumference, BMI, or waist-to-height ratio (Supplementary

Table S2).

The findings were insensitive to the dual use of HbA1c as an eligibility criterion and an adjustment covariate. When continuous HbA1c was removed from the model while retaining the HbA1c < 5.7% eligibility criterion, the HOMA-IR-by-age interaction was essentially unchanged,

$$F(2, 40) = 5.57, \quad p = 0.00732.$$

The corresponding HOMA-IR coefficients were 0.0284, 0.0521, and -0.0171 among adults aged 20–39, 40–54, and 55 years or older, respectively, compared with 0.0266, 0.0511, and -0.0189 in the primary model.

The age interaction also persisted when the HbA1c eligibility criterion was progressively relaxed. The interaction yielded

$$F(2, 40) = 5.74, \quad p = 0.00644$$

for HbA1c < 6.0%;

$$F(2, 40) = 5.58, \quad p = 0.00731$$

for HbA1c < 6.5%; and

$$F(2, 40) = 5.74, \quad p = 0.00643$$

when the HbA1c eligibility threshold was removed while participants with diagnosed diabetes remained excluded. These analyses indicate that the observed age modification was not explained solely by selection at the 5.7% HbA1c threshold.

Alternative continuous-age specifications provided complementary evidence. As reported above, the linear interaction yielded $p = 0.0544$, the joint quadratic interaction yielded $p = 0.0421$, and the four-degree-of-freedom natural cubic spline interaction yielded $p = 0.0262$, indicating that evidence for age modification was stronger when nonlinear variation across adulthood was permitted.

The expanded combined-period analysis additionally tested whether the age-dependent HOMA-IR association differed between the 2015–2016 and 2017–March 2020 components. For the primary log-hs-CRP specification, the period heterogeneity test yielded

$$F(2, 40) = 0.728, \quad p = 0.489,$$

providing no strong evidence of between-period heterogeneity in the age-dependent association.

Finally, leave-one-survey-cluster-out analysis generated 79 refitted models after sequential removal of each unique masked stratum–PSU combination. Across these refits, the HOMA-IR-by-age interaction F statistic ranged from 3.59 to 7.66, with corresponding p -values ranging from 0.00157 to 0.0369. No single masked stratum–PSU cluster eliminated evidence of the interaction.

Additional supportive analyses conducted during the methodological audit evaluated influential observations, smoking, medication burden, statin use, renal and liver/metabolic biochemical variables, expanded covariate sets, and sex heterogeneity. These exploratory analyses did not identify an obvious single factor accounting for the overall age-dependent pattern and were not treated as additional primary hypothesis tests.

3.6 Use of Generative Artificial Intelligence

Generative artificial intelligence (ChatGPT, OpenAI) was used extensively as a research-support tool during this project, including assistance with drafting and revising portions of the manuscript and with writing, optimizing, and debugging portions of the reproducible Python analysis pipeline. The authors determined the research questions, analytic strategy, statistical methods, interpretation of results, and conclusions. AI-assisted analytical code and outputs were subject to human review and validation against the prespecified analytic framework, mathematical expectations, and NHANES survey-design constraints. The primary analysis was additionally reproduced independently in R as a cross-software validation. All manuscript content was critically reviewed and approved by the authors, who take full responsibility for the accuracy, integrity, interpretation, and conclusions of the work.

4 Discussion

4.1 Principal Findings

In this nationally representative analysis of U.S. adults without diagnosed diabetes and with HbA1c below 5.7%, HOMA-IR was associated with systemic inflammatory status in an age-dependent manner. In the primary 2015–March 2020 analysis, the adjusted association between HOMA-IR and log-transformed hs-CRP differed across age groups. Higher HOMA-IR was associated with higher hs-CRP among adults aged 20–39 years and more strongly among those aged 40–54 years, whereas the corresponding association among adults aged 55 years or older was attenuated and statistically indistinguishable from zero.

Importantly, the evidence for age modification arose from a formal interaction test rather than from comparison of statistical significance across separately fitted age-stratified models. The age-

dependent pattern was statistically compatible across the independent 2015–2016 and 2017–2018 NHANES cycles, and continuous-age modeling produced a broadly consistent pattern in which the estimated HOMA-IR association remained positive through early and middle adulthood and weakened later in adulthood.

The qualitative age-dependent pattern was generally recognizable across multiple sensitivity analyses addressing outcome transformation, inflammatory extremes, exposure scale, adiposity specification, HbA1c eligibility, additional covariates, and influential survey clusters. However, the strength of formal interaction evidence varied across specifications, particularly when HOMA-IR was log-transformed or when log-transformed hs-CRP was analyzed after restriction to $\text{hs-CRP} \leq 10 \text{ mg/L}$. The overall evidence therefore supports age-dependent heterogeneity in the HOMA-IR–hs-CRP association while also indicating that the estimated strength of effect modification is sensitive to modeling scale.

4.2 Insulin Resistance and Inflammation Below Conventional Glycemic Thresholds

The association between insulin resistance and systemic inflammation is not itself a novel observation. Previous population-based studies, including analyses of U.S. NHANES data and other national and longitudinal cohorts, have reported associations between HOMA-IR or fasting insulin and C-reactive protein in adults without diabetes [Chen et al., 2004, Meng et al., 2007, Park et al., 2009, Missel et al., 2021]. Related studies have also demonstrated substantial metabolic heterogeneity among individuals without overt dysglycemia [Caporaso et al., 2020].

The present study addresses a narrower question. Rather than asking simply whether insulin resistance is associated with inflammation, we restricted the primary population to adults remaining below the conventional HbA1c prediabetes threshold of 5.7% [American Diabetes Association Professional Practice Committee, 2026] and examined whether the strength of the HOMA-IR–hs-CRP association itself varied across adulthood. Prior work has examined demographic heterogeneity in insulin resistance–inflammation associations, including nationally representative studies evaluating sex and age [Caporaso et al., 2020, Shahid et al., 2023]. Comparatively less evidence, however, has focused specifically on formal age modification of this relationship within a population remaining below the conventional HbA1c prediabetes threshold.

This distinction matters because HbA1c and HOMA-IR characterize different aspects of metabolic physiology. HbA1c summarizes glycemic exposure [Nathan et al., 2007, American Diabetes Association Professional Practice Committee, 2026], whereas HOMA-IR estimates insulin resistance from the fasting glucose–insulin relationship [Matthews et al., 1985, Wallace et al., 2004]. During compensatory stages of insulin resistance, increased insulin secretion may contribute to maintenance of glucose concentrations despite diminished insulin sensitivity [Rodzeń et al., 2026]. Consequently, similar HbA1c values need not imply similar insulin dynamics.

Consistent with this distinction, HOMA-IR values in the present analytic population were not concentrated at a single level despite restriction to HbA1c below 5.7%. Moreover, survey-weighted median HOMA-IR was relatively similar across the three age groups, even though the estimated relationship between HOMA-IR and hs-CRP differed substantially by age. The age-dependent association therefore should not be interpreted simply as a consequence of middle-aged adults having higher average HOMA-IR than the other age groups.

The present findings are thus consistent with meaningful metabolic heterogeneity within an HbA1c-defined population that remains below the prediabetes threshold. They do not establish that such heterogeneity is pathological in every individual, but they indicate that glycemic classification alone does not fully characterize variation in insulin resistance or its relationship with systemic inflammatory status.

4.3 Age Modification

Age modification was not the original hypothesis motivating this study. The initial scientific question concerned whether HOMA-IR identified inflammatory heterogeneity among adults remaining below the conventional HbA1c prediabetes threshold. The age-dependent pattern emerged during evaluation of the initial stratified analyses and was subsequently subjected to formal interaction testing, independent-cycle replication, continuous-age modeling, and sensitivity analyses. Accordingly, the specific shape of the age-dependent association should be interpreted as an empirically identified and subsequently robustness-tested finding rather than as a prospectively specified biological prediction.

Age modification is a central finding of the present analysis. In the primary model, the HOMA-IR–hs-CRP association was positive in younger adults, strongest in the 40–54-year group, and attenuated among adults aged 55 years or older. Continuous-age modeling provided additional evidence that the phenomenon is not well described by a constant HOMA-IR effect across the adult lifespan.

This pattern should not be interpreted as evidence that hs-CRP declines with age. The interaction describes how the adjusted association between HOMA-IR and hs-CRP changes with age, not how hs-CRP itself changes with age. Indeed, median hs-CRP was modestly higher in the oldest age group. Nor should the spline be interpreted as identifying a discrete biological age at which the association disappears. The continuous-age results are more consistent with gradual variation in the strength of the association across adulthood.

The mechanism underlying the older-age attenuation cannot be determined from the present study. One possibility is that the relative contribution of insulin resistance to systemic inflammatory variation changes as inflammatory, adipose, metabolic, and cellular processes evolve with age [Park et al., 2014]. Medication exposure, comorbidity burden, lifestyle changes, and other determinants of inflammatory status may add further heterogeneity later in life, although the present study

cannot determine the contribution of these factors. Such processes could reduce the extent to which variation in HOMA-IR tracks variation in hs-CRP even if overall inflammatory burden remains elevated.

Selection may also contribute. Because the present analysis intentionally excludes individuals with diagnosed diabetes and those with HbA1c at or above 5.7%, the meaning of remaining in the analytic population changes with age. An older participant who continues to satisfy these criteria may differ systematically from a peer who previously crossed a glycemic threshold or developed diagnosed diabetes and was therefore excluded.

However, sensitivity analyses progressively relaxing the HbA1c eligibility criterion to $< 6.0\%$, $< 6.5\%$, and ultimately no HbA1c cutoff did not restore a positive older-adult HOMA-IR–hs-CRP association. This finding argues against a simple explanation in which the older-age attenuation is produced solely by exclusion at the 5.7% HbA1c threshold. It does not, however, eliminate broader selection effects or distinguish them from biological effect modification, treatment effects, cohort differences, behavioral adaptation, residual confounding, or combinations of these processes.

The data therefore establish attenuation of an association; they do not establish why that attenuation occurs. Similarly, the older-adult point estimate should not be interpreted as evidence of a protective relationship between insulin resistance and inflammation.

The role of adiposity warrants particular consideration. In unadjusted analyses, HOMA-IR was positively associated with hs-CRP across all three age groups, whereas adjustment for waist circumference accounted for most of the transition to the age-dependent pattern observed in the primary model. Importantly, however, this pattern was not specific to waist circumference. In common-sample analyses spanning the full study period, substitution of either BMI or waist-to-height ratio for waist circumference produced the same qualitative pattern of positive associations in early and middle adulthood and marked attenuation in older adults. Thus, the observed age-dependent pattern appears to depend importantly on accounting for adiposity, but not on a single full-period anthropometric measure.

These findings should not be interpreted as showing that adiposity simply confounds an otherwise causal association between insulin resistance and inflammation. Central adiposity, insulin resistance, and inflammatory signaling are biologically interrelated, and their temporal and causal relationships cannot be resolved in cross-sectional data. Waist circumference may therefore represent an upstream determinant, an intermediate phenotype, a correlate of shared metabolic processes, or some combination of these roles. The adjusted estimates are consequently best interpreted as conditional associations describing variation in inflammatory status associated with HOMA-IR among individuals with similar measured adiposity and other model covariates.

In the restricted survey period in which waist-to-hip ratio was available, the older-age association was less attenuated after adjustment for waist-to-hip ratio than after adjustment for the other

anthropometric measures; however, evidence for age modification was generally weaker in this restricted sample even when waist circumference was retained.

The observed age modification was also insensitive to whether continuous HbA1c was retained as an adjustment covariate after restriction to HbA1c < 5.7%. Removing continuous HbA1c produced age-specific coefficients that were nearly identical to those in the primary model, indicating that the principal finding was not driven by the dual role of HbA1c in population definition and covariate adjustment.

4.4 Reconsidering the Metabolic “Blind Spot”

The results permit a more precise formulation of the metabolic “blind spot” that motivated this study. We do not use the term to suggest that HbA1c is an inadequate measure of glycemia, nor do the findings imply that individuals with HbA1c below 5.7% should be considered metabolically abnormal solely because they have a higher HOMA-IR.

Rather, the blind spot is one of *measurement domain*. HbA1c answers a question about glycemic exposure [Nathan et al., 2007, American Diabetes Association Professional Practice Committee, 2026]. It does not directly measure the amount of insulin required to maintain that glycemic state or the degree of tissue insulin sensitivity [Matthews et al., 1985, Wallace et al., 2004]. Two individuals can therefore occupy the same HbA1c-defined category while differing substantially in fasting insulin and HOMA-IR.

The original hypothesis of the study was therefore supported in a qualified rather than universal sense. Among adults remaining below the conventional HbA1c prediabetes threshold, variation in HOMA-IR was associated with variation in systemic inflammatory status. However, this relationship was not uniform across adulthood. The evidence was strongest in early and middle adulthood and attenuated in the older group.

This refinement is important. The relevant inference is not that HbA1c “misses disease,” but that an HbA1c value below 5.7% does not define a metabolically homogeneous population. HOMA-IR captures variation in insulin dynamics that is not represented directly by HbA1c, and at least some of that variation is associated with systemic inflammatory status.

The distinction may be more useful than framing HOMA-IR and HbA1c as competing biomarkers. They measure different features of metabolic regulation. The clinically relevant question is therefore not whether HOMA-IR should replace HbA1c, but whether direct information about insulin resistance adds useful predictive or decision-relevant information to conventional glycemic assessment in selected populations.

4.5 Clinical and Public-Health Implications

Recent reviews have called for improved recognition and characterization of insulin resistance before the development of overt metabolic disease and have emphasized the absence of standardized criteria for its routine clinical identification [Rodzeń et al., 2026]. The present findings are relevant to that broader discussion because they demonstrate measurable heterogeneity in insulin resistance and its inflammatory correlates among adults who remain below the conventional HbA1c prediabetes threshold.

The present study does not, however, establish the clinical utility of routine fasting insulin measurement, and our findings alone do not justify changing screening recommendations. A cross-sectional association with hs-CRP is insufficient to demonstrate that measuring HOMA-IR improves prediction, alters clinical decision-making, or ultimately improves patient outcomes.

The findings instead provide a rationale for prospective evaluation of those questions. Future studies could test whether incorporating fasting insulin or validated measures of insulin resistance into risk assessment improves prediction beyond HbA1c, fasting glucose, anthropometry, lipids, and established clinical risk factors. Longitudinal studies would be particularly valuable for clarifying temporal relationships between insulin resistance, inflammation, and subsequent metabolic risk [Park et al., 2009, Yan et al., 2019], including whether individuals with higher HOMA-IR but HbA1c below 5.7% are at increased risk of subsequent dysglycemia, cardiovascular events, metabolic liver disease, or other clinically relevant outcomes.

Such work should also evaluate whether predictive utility varies by age. The present analysis suggests that the relationship between insulin resistance and inflammatory status is not uniform across adulthood. Accordingly, even if additional insulin-resistance measurement ultimately proves clinically informative, there is no basis for assuming that the same threshold, interpretation, or screening strategy would necessarily perform equivalently across age groups.

This perspective is consistent with recent calls for improved diagnostic frameworks for insulin resistance while also emphasizing the need for empirical validation before policy or screening recommendations are changed. The present study contributes population-level evidence relevant to that discussion but does not resolve the clinical-policy question.

4.6 Strengths and Limitations

This study has several strengths. The primary analysis uses a large, nationally representative pre-pandemic NHANES population and incorporates the fasting-subsample weights, stratification, and clustering required for design-based inference. The age-dependent pattern identified during methodological evaluation was subsequently tested using a formal HOMA-IR-by-age interaction rather than by comparing subgroup p -values.

The analysis also provides several complementary assessments of reproducibility and robustness. The age-dependent pattern was evaluated across independent conventional NHANES cycles, examined continuously using linear, quadratic, and spline-based age specifications, and subjected to sensitivity analyses addressing outcome and exposure scale, inflammatory extremes, adiposity adjustment, HbA1c eligibility, and survey-cluster influence. No single masked stratum-PSU cluster eliminated evidence of the primary interaction. Importantly, not every alternative specification produced equally strong formal interaction evidence, and these results are reported rather than excluded from consideration.

Several limitations are equally important. First, NHANES is cross-sectional; temporal ordering between insulin resistance and inflammation cannot be established. The observed association therefore cannot determine whether insulin resistance contributes to inflammation, inflammation contributes to insulin resistance, both arise from shared upstream processes, or some combination of these mechanisms operates.

Second, HOMA-IR is a surrogate measure derived from fasting glucose and insulin rather than a direct measurement of insulin sensitivity such as the hyperinsulinemic-euglycemic clamp [DeFronzo et al., 1979, Wallace et al., 2004]. Although it is widely used in population research and correlates with more intensive physiological measures, measurement error and biological variability remain.

Third, hs-CRP is a nonspecific marker of systemic inflammation and can be influenced by infection, chronic inflammatory conditions, adiposity, medication use, and other factors unrelated to insulin resistance [Pepys and Hirschfield, 2003]. Sensitivity analyses limiting extreme hs-CRP values and adjusting for additional available covariates reduce but cannot eliminate this concern.

Fourth, formal evidence for effect modification varied across model specifications. Although the qualitative pattern of positive associations earlier in adulthood and attenuation later in adulthood was generally recognizable, interaction tests were weaker under some exposure and outcome transformations. The magnitude and statistical strength of age modification should therefore not be regarded as invariant to modeling scale.

Fifth, residual and unmeasured confounding remain possible despite adjustment for HbA1c, adiposity, sex, age, and non-HDL cholesterol and the additional sensitivity analyses. Although the age-dependent pattern was reproduced using multiple anthropometric specifications, these measures are imperfect proxies for body composition and fat distribution and cannot fully account for the complex relationship between adiposity, insulin resistance, and inflammation.

Sixth, the primary analysis used complete cases, creating the possibility of selection bias if missingness was related to the exposure, outcome, or covariates under study. However, only 103 of 3,182 participants (3.2%) who otherwise met the eligibility and measurement requirements were excluded because of incomplete primary-model covariate data. The small proportion of exclusions limits, but does not eliminate, this concern.

Seventh, restricting the primary study population to participants without diagnosed diabetes and with HbA1c below 5.7% creates a deliberately selected analytic population. This restriction is essential to the scientific question but complicates interpretation of age differences because older adults remaining below the threshold may not be directly comparable, in a life-course sense, with younger adults meeting the same criteria.

The use of NHANES probability sampling and appropriate fasting-subsample weights supports population-level inference to the noninstitutionalized U.S. adult population meeting the study eligibility criteria. The findings should not, however, be assumed to generalize to institutionalized populations, populations outside the United States, individuals with diagnosed diabetes or HbA1c $\geq 5.7\%$, or populations whose metabolic and inflammatory characteristics differ substantially from those represented in NHANES.

Finally, age modification was identified during methodological evaluation rather than prospectively specified as the original study hypothesis. Although the resulting pattern was subsequently subjected to formal interaction testing, independent-cycle replication, continuous-age modeling, and extensive sensitivity analysis, the precise age-dependent shape should be regarded as a subsequently validated empirical observation rather than a prospectively predicted biological pattern. The categorical results likewise should not be interpreted as establishing a biological threshold at age 55.

4.7 Conclusion

Among U.S. adults without diagnosed diabetes and with HbA1c below 5.7%, the adjusted association between HOMA-IR and systemic inflammatory status varied across age. Higher HOMA-IR was associated with higher hs-CRP in young and middle-aged adults, whereas the corresponding association was attenuated among older adults. The age-dependent pattern was supported by formal interaction testing, independent-cycle analyses, continuous-age modeling, and multiple robustness analyses, although the strength of formal interaction evidence varied across model specifications.

These findings support a qualified version of the original metabolic “blind spot” hypothesis: adults remaining below the conventional HbA1c prediabetes threshold are not metabolically homogeneous, and variation in insulin resistance can be associated with inflammatory physiology not fully characterized by glycemic classification alone. The present data do not establish whether this additional information improves prospective risk prediction or clinical outcomes. Determining whether direct measurement of insulin resistance adds clinically useful information to conventional glycemic assessment, and whether that utility itself varies with age, requires prospective investigation.

Author Contributions

S.M. conceived the study, developed the analytic framework, conducted the statistical analyses, interpreted the results, and drafted the manuscript. J.B. contributed to study development, interpretation of the findings, and critical revision of the manuscript. Both authors reviewed and approved the final manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Ethics Statement

This study used publicly available, de-identified data from the National Health and Nutrition Examination Survey (NHANES). NHANES protocols were approved by the National Center for Health Statistics Research Ethics Review Board, and written informed consent was obtained from participants in accordance with NHANES procedures. Because the present study involved secondary analysis of publicly available, de-identified data, no additional institutional review board approval or informed consent was required.

Data Availability

The NHANES data used in this study are publicly available from the National Center for Health Statistics.

Code Availability

The computational code, aggregate analysis outputs, and independent cross-software validation supporting this study are archived on Zenodo at DOI: [10.5281/zenodo.23176026](https://doi.org/10.5281/zenodo.23176026). The computational repository is also publicly available through GitHub.

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